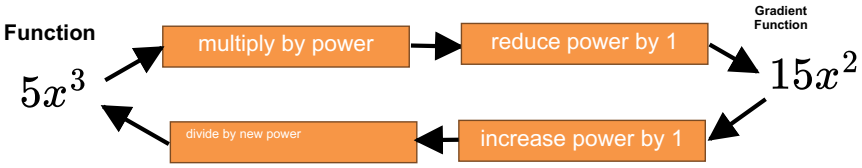
1.36 Integration: Indefinite Integrals

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Integrating x^n terms

3

Integration is the **opposite of differentiation**.
(For this reason it is also called '*antidifferentiation*')



However, there's one added complication…

Find y when $\frac{dy}{dx} = 3x^2$

c is known as a **constant of integration**

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Examples

4

Find y when:

$\frac{dy}{dx} = 4x^3 \qquad y =$

$\frac{dy}{dx} = x^5 \qquad y =$

$\frac{dy}{dx} = 3x^{1/2} \qquad y =$

Textbook Error on Pg289: "When integrating polynomials", but Example 3 is not a polynomial because it has negative and fractional powers.

1.36 Integration: Indefinite Integrals

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More Fractional Examples

5

Find y when:

$\frac{dy}{dx} = \frac{4}{\sqrt{x}}$ $y =$

$\frac{dy}{dx} = 5x^{-2}$ $y =$

$\frac{dy}{dx} = 4x^{2/3}$ $y =$

$\frac{dy}{dx} = 10x^{-2/7}$ $y =$

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Test Your Understanding

6

Find $f(x)$ when:

$\mathrm{f}'(x) = 2x + 7$ $\mathrm{f}(x) =$

$\mathrm{f}'(x) = x^2 - 1$ $\mathrm{f}(x) =$

$\mathrm{f}'(x) = \frac{2}{x^7}$ $\mathrm{f}(x) =$

$\mathrm{f}'(x) = \sqrt[3]{x}$ $\mathrm{f}(x) =$

$\mathrm{f}'(x) = 33x^{5/6}$ $\mathrm{f}(x) =$

Note: In case you're wondering what happens if $\frac{dy}{dx} = \frac{1}{x} = x^{-1}$, the problem is that after adding 1 to the power, we'd be dividing by 0. You will learn how to integrate $\frac{1}{x}$ in Year 2.

1.36 Integration: Indefinite Integrals

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Exercise 13A – Y1 pages 289-290

7

4. Find y given that $\frac{dy}{dx} = (2x+3)^2$

5. Find $f(x)$ given that
$f'(x) = 3x^{-2} + 6x^{\frac{1}{2}} + x - 4$

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Exercise 13A – Y1 pages 290

8

Challenge. Find y given that $\frac{dy}{dx} = \left(2\sqrt{x} - x^2\right)\left(\frac{3+x}{x^5}\right)$

1.36 Integration: Indefinite Integrals

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Integration notation

9

The following notation could be used to differentiate an expression:

$$\frac{d}{dx}\left(5x^2\right) = 10x$$

The dx here means differentiating "with respect to x".

There is similarly notation for integrating an expression:

$$\int 10x \ \ dx = 5x^2 + c$$

"Integrate…"

"…this expression"

"…with respect to x"
(the dx is needed just as it was needed in the differentiation notation at the top of this slide)

This is known as **indefinite integration**, in contrast to definite integration, which we'll see later in the chapter.

It is called 'indefinite' because the exact expression is unknown (due to the $+c$).

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Examples

10

Note: The brackets are required if there's multiple terms.

Find $\int (x^{-\frac{3}{2}} + 2)dx$

Find $\int (6t^2 - 1)dt$

Note the dt instead of dx.

Find $\int (px^3 + q)dx$ where p and q are constants.

1.36 Integration: Indefinite Integrals

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Test Your Understanding

11

Edexcel C1 May 2014(R) Q4b

Given that $y = 2x^5 + \frac{6}{\sqrt{x}}$, $x > 0$, find in their simplest form

(b) $\int y \, \mathrm{d}x$ \hfill (3)

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Exercise 13B – Y1 pages 291-293

12

10 Given that $\mathrm{f}(x) = \frac{6}{x^2} + 4\sqrt{x} - 3x + 2$, $x > 0$, find $\int \mathrm{f}(x)\mathrm{d}x$.

11 Find $\int \left(8x^3 + 6x - \frac{3}{\sqrt{x}}\right)\mathrm{d}x$, giving each term in its simplest form.

1.36 Integration: Indefinite Integrals

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Exercise 13B – Y1 pages 291-293

13

12 a Show that $(2+5\sqrt{x})^2$ can be written as $4+k\sqrt{x}+25x$, where k is a constant to be found.

b Hence find $\int (2+5\sqrt{x})^2 \, \mathrm{d}x$.

13 Given that $y = 3x^5 - \frac{4}{\sqrt{x}}$, $x > 0$, find $\int y \, \mathrm{d}x$ in its simplest form.

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Exercise 13B – Y1 pages 291-293

14

14 $\int \left(\frac{p}{2x^2} + pq\right) \mathrm{d}x = \frac{2}{x} + 10x + c$ Find the value of p and the value of q.

15 $\mathrm{f}(x) = (2-x)^{10}$.

Given that x is small, and so terms in x^3 and higher powers of x can be ignored:

a find an approximation for $\mathrm{f}(x)$ in the form $A + Bx + Cx^2$

b find an approximation for $\int \mathrm{f}(x) \, \mathrm{d}x$.

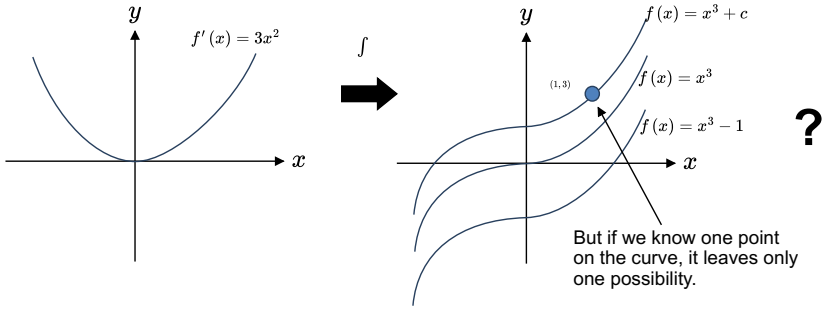
1.36 Integration: Indefinite Integrals

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Finding constant of integration

15

Recall that when we integrate, we get a constant of integration, which could be any real value. This means **we don't know what the exact original function was**.



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Test Your Understanding

16

Edexcel C1 May 2014 Q10

A curve with equation $y = \mathrm{f}(x)$ passes through the point $(4, 25)$.

Given that
$\mathrm{f}'(x) = \frac{3}{8}x^2 - 10x^{-1/2} + 1, \qquad x > 0$

(a) find $\mathrm{f}(x)$, simplifying each term. \hfill (5)

(b) Find an equation of the normal to the curve at the point $(4, 25)$.

Give your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.
\hfill (5)

(b)

1.36 Integration: Indefinite Integrals

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Exercise 13C – Y1 pages 294-295

17

4 The curve with equation $y = \mathrm{f}(x)$ passes through the point $(-1, 0)$. Given that $\mathrm{f}'(x) = 9x^2 + 4x - 3$, find $\mathrm{f}(x)$.

5 $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 3x^{-1/2} - 2x\sqrt{x}$, $x > 0$.
Given that $y = 10$ at $x = 4$, find y in terms of x, giving each term in its simplest form.

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Exercise 13C – Y1 pages 294-295

18

6 Given that $\dfrac{6x + 5x^{3/2}}{\sqrt{x}}$ can be written in the form $6x^p + 5x^q$,

a write down the value of p and the value of q.

Given that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{6x + 5x^{3/2}}{\sqrt{x}}$ and that $y = 100$ when $x = 9$,

b find y in terms of x, simplifying the coefficient of each term.